



Technical Service Bulletin: TSB200196

Released Date: 10-Sep-2025

Updated Turbocharger with CSeal and New Bearing Housing

## Updated Turbocharger with CSeal and New Bearing Housing

### Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

### Contents

#### Product Affected

- QSK19 CM500
- QSK19 CM850 MCRS
- QSK19 CM2150 MCRS
- QSK23 CM500
- QSK60 CM2150
- QSK60 CM2350 K116
- QSK60 CM2350 K117
- QSK60 CM2350 K121
- QSK60 CM2350 K136
- QSK60 CM500
- QSK78 CM500

#### Description of Change

The updated turbocharger has a new bearing housing and a revised bearing housing to turbine housing joint.

The bearing housing has been redesigned to allow for better oil flow to resolve compressor side oil leaks.

The turbine to bearing housing joint incorporates a revised geometry, CSeal, and new capscrews. The redesigned joint has been developed to resolve capscrew fractures in the joint. The new capscrews are longer than previous and made from B16 material. The CSeal is carried over from a previous product improvement which has shown to significantly reduce capscrew failures in the turbine/bearing housing joint. The revised joint geometry includes a thicker bearing housing flange to increase bolt preload and the capscrew torque increased to 25 N•m [ 18.44 ftlb ].

### Reason for Change

The changes have been made to address component malfunctions during operation.

### Customer Communication

The improvements mentioned above are due to be rolled out across the majority of High Horsepower engines which use turbochargers with a bolted turbine to bearing housing joint.

As these products get updated, this TSB will be updated with the new part and service kit numbers.

### Service Parts Availability

Service parts are available. As future turbochargers get improvement updates, Table 1 will be updated with new part numbers.

**Table 1, Turbocharger and Service Part Number**

| Part Description | Engine Platform             | Existing Part number                        | Existing Service Kit Number | Obsolete | Superseded | New Part Number                             | New Service Kit Number |
|------------------|-----------------------------|---------------------------------------------|-----------------------------|----------|------------|---------------------------------------------|------------------------|
| Turbocharger     | QSK19 CM500                 | 4033047, 4037424                            | 4089964                     | No       | No         | 6391532, 6391531                            | 6391540                |
| Turbocharger     | QSK19 CM850 and CM2150 MCRS | 4040240, 3777194, 4033010, 4040241, 5552117 | 4956137                     | No       | No         | 6390248, 6390246, 6390250, 6390249, 6390247 | 6390251                |
| Turbocharger     | QSK23 CM500                 | 3798613                                     | 5321707                     | Yes      | Yes        | 6390477                                     | 6390867                |
| Turbocharger     | QSK60 K117 and K136         | 5458891                                     | 5460081                     | Yes      | Yes        | 5603061                                     | 5603065                |
|                  |                             | 5548275                                     |                             |          |            |                                             |                        |
| Turbocharger     | QSK60 CM2150                | 4047296                                     | 5321637                     | Yes      | Yes        | 5638766                                     | 5638753                |
|                  |                             | 4047297                                     | 5321619                     | Yes      | Yes        | 5638770                                     | 5638754                |
| Turbocharger     | QSK78 CM500 Single Stage    | 3786518, 3792044, 3792045                   | 5633303                     | Yes      | Yes        | 5638761, 5638762, 5638763                   | 5638751                |
| Turbocharger     | QSK60 CM500 2stage          | 3594185, 3594186, 4033019                   | 3800499                     | Yes      | Yes        | 5638758, 5638759, 5638760                   | 5638750                |

| Table 1, Turbocharger and Service Part Number |                            |                           |                             |          |            |                           |                        |
|-----------------------------------------------|----------------------------|---------------------------|-----------------------------|----------|------------|---------------------------|------------------------|
| Part Description                              | Engine Platform            | Existing Part number      | Existing Service Kit Number | Obsolete | Superseded | New Part Number           | New Service Kit Number |
| Turbocharger                                  | QSK60 CM500 2stage         | 2843029, 2843030, 4033441 | 2882097                     | Yes      | Yes        | 5638755, 5638756, 5638757 | 5638749                |
| Turbocharger                                  | QSK60 CM2350 K121          | 5352262                   | 5352263                     | Yes      | Yes        | 5638765                   | 5638752                |
| Turbocharger                                  | QSK60 CM2350 K116 and K136 | 5458892, 5458893, 5460080 | 5460081                     | Yes      | Yes        | 5603062, 5603063, 5603064 | 5603065                |

### Part Inventory Action

Use existing part number inventory before using the new part number.

### Production Status

Implemented for production.

| Table 2, Production Information  |                             |              |                          |                       |
|----------------------------------|-----------------------------|--------------|--------------------------|-----------------------|
| ESN First                        | Build Date <sup>1</sup>     | Engine Model | Turbocharger Part Number | Plant                 |
| 37293715<br>60608152<br>(ReCon®) | 13 May 2024<br>17 May 2024  | QSK19 CM500  | 6391532                  | Seymour Engine Plant  |
| 37292361                         | 18 December 2023            | QSK19 CM2150 | 6390250                  | Seymour Engine Plant  |
| 37293696<br>60608174<br>(ReCon®) | 09 May 2024<br>25 June 2024 | QSK19 CM2150 | 6390246                  | Seymour Engine Plant  |
| 85016741                         | 18 August 2023              | QSK23 CM500  | 6390477                  | PHP                   |
| 33221662                         | 18 May 2020                 | QSK60 K136   | 5603061                  | Daventry Engine Plant |
| 33223224                         | 2 November 2020             | QSK60 CM2150 | 5638766, 5638770         | Daventry Engine Plant |
| 33224739                         | 7 May 2021                  | QSK60 K135   | 5638764                  | Daventry Engine Plant |

| Table 2, Production Information                                      |                         |                   |                          |                       |
|----------------------------------------------------------------------|-------------------------|-------------------|--------------------------|-----------------------|
| ESN First                                                            | Build Date <sup>1</sup> | Engine Model      | Turbocharger Part Number | Plant                 |
| 66306346                                                             | 24 November 2021        | QSK78 CM500       | 5638762,<br>5638763      | Daventry Engine Plant |
| 33224816                                                             | 17 May 2021             | QSK60 CM500       | 5638758,<br>5638759      | Daventry Engine Plant |
| 33224816                                                             | 17 May 2020             | QSK60 CM500       | 5638755,<br>5638755      | Daventry Engine Plant |
| 33221662                                                             | 18 May 2020             | QSK60 CM2350 K136 | 5603062,<br>5603063      | Daventry Engine Plant |
| <sup>1</sup> Engine build date can be found on the engine dataplate. |                         |                   |                          |                       |

## Document History

| Date      | Details                                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------------------|
| 2020-9-25 | Module Created                                                                                                   |
| 2021-1-26 | Added QSK60 CM2150.                                                                                              |
| 2021-6-24 | Updated Product Affected section. Updated Table 1.                                                               |
| 2021-9-10 | Added third row to Table 2.                                                                                      |
| 2022-6-22 | Updated Table 2.                                                                                                 |
| 2023-9-26 | Added QSK23 CM500. Updated Table 1 and Table 2.                                                                  |
| 2025-9-5  | Updated Product Affected with QSK19 CM500, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS. Updated Table 1 and Table 2. |